

How the Proof Works

We are asking whether a viscous fluid that starts smooth can ever stop being smooth. This proof answers no. The difficulty is that viscosity is not the only part of the equation changing the fluid. The fluid also carries and deforms its own velocity:

$$\underbrace{\partial_t u}_{\text{local acceleration}} + \underbrace{(u \cdot \nabla) u}_{\text{nonlinear transport}} = - \underbrace{\nabla p}_{\text{pressure}} + \underbrace{\nu \Delta u}_{\text{viscous diffusion}}, \quad \nabla \cdot u = 0.$$

The nonlinear term can sharpen the flow while viscosity smooths it. Pressure keeps the motion incompressible and couples every local change to the surrounding field. The regularity problem is decided by whether nonlinear sharpening can outrun viscous smoothing before some finite time T_* .

The place to see that danger is vortex stretching. Write

$$\omega := \nabla \times u, \quad S := \frac{1}{2}(\nabla u + (\nabla u)^\top).$$

Taking the curl of Navier–Stokes gives

$$(\partial_t + u \cdot \nabla)\omega = S\omega + \nu \Delta \omega.$$

When the vorticity points along an extensional direction of S , the term $S\omega$ strengthens it. The vortex stretches lengthwise; incompressibility forces its cross-section to narrow; and the velocity turns through a larger change across a smaller radius. Viscosity acts against that sharpening through $\nu \Delta \omega$. After integration over space, the contest is visible in one line:

$$\frac{1}{2} \frac{d}{dt} \|\omega(t)\|_2^2 = \int_{\mathbb{R}^3} \omega \cdot S\omega \, dx - \nu \|\nabla \omega(t)\|_2^2.$$

The first term is nonlinear production. The second is viscous removal.

Nothing here requires the kinetic energy itself to become infinite. In fact,

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \nu \|\nabla u(t)\|_2^2 = 0.$$

The total energy decreases. The unresolved possibility is concentration: finite energy can be carried by thinner regions while the velocity gradients inside those regions grow. A vortex can be losing energy and still be sharpening near its core.

So we shrink the picture and ask whether the balance changes. For $\lambda > 0$, set

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

Every part of the equation picks up one common factor:

$$\begin{aligned} \partial_t u_\lambda &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), & (u_\lambda \cdot \nabla) u_\lambda &= \lambda^3 ((u \cdot \nabla) u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), & \nu \Delta u_\lambda &= \lambda^3 \nu (\Delta u)(\lambda x, \lambda^2 t). \end{aligned}$$

Viscosity does not acquire an extra power of λ that makes it win automatically in a smaller vortex. Nonlinear transport, pressure, local acceleration, and diffusion remain matched by scaling.

The corresponding Sobolev calculation tells us where that match lives. Since

$$\|u_\lambda(\cdot, t)\|_{\dot{H}^s} = \lambda^{s-1/2} \|u(\cdot, \lambda^2 t)\|_{\dot{H}^s},$$

concentration lowers the norm below $s = \frac{1}{2}$, raises it above $s = \frac{1}{2}$, and leaves it unchanged at

$$s = \frac{1}{2}.$$

This is the critical height. It is not a claim that nonlinear production and viscous dissipation are numerically equal at each moment. It says that scale alone gives neither one an advantage there.

Now keep following the vortex as its core narrows. If its radius is ℓ , a second spatial derivative across that core has size ℓ^{-2} relative to the velocity change it sees. The viscous term acts on the corresponding interval

$$|\nu \Delta u| \sim \frac{\nu}{\ell^2} |u|, \quad \tau_\ell \sim \frac{\ell^2}{\nu}.$$

If the core kept narrowing through $\ell_j = \rho^j \ell_0$ with $0 < \rho < 1$, those intervals would satisfy

$$\sum_{j=0}^{\infty} \frac{\ell_j^2}{\nu} = \frac{\ell_0^2}{\nu} \sum_{j=0}^{\infty} \rho^{2j} = \frac{\ell_0^2}{\nu(1-\rho^2)} < \infty.$$

That is the Zeno picture inside the vortex: each thinner core asks the equation to respond faster, and the resulting intervals can fit below one finite terminal time. It does not assert that a solution follows this cascade. It shows the failure pattern the proof has to prevent.

The vortex picture has now reached a finite endpoint. The derivative tower begins by asking what an unbounded change in velocity would have to do in the time left. For every finite temporal order j ,

$$\partial_t^j u(t) - \partial_t^j u(0) = \int_0^t \partial_s^{j+1} u(s) ds.$$

If $\partial_t^{j+1} u$ stayed bounded on a finite interval, then $\partial_t^j u$ could not become unbounded there. Apply that observation from velocity to each response that follows it.

In order for a singularity to form, velocity has to blow up. In order for that to happen, its acceleration would also have to blow up, which means its jerk has to blow up, and so on, and so on:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \dots$$

This is the turn in the argument. Acceleration is already written into Navier–Stokes. Once we substitute the equation for it and differentiate, the jerk appears. Differentiate again and the next response appears. The first three rungs are

$$\begin{aligned} \partial_t u &= \nu \Delta u - (u \cdot \nabla) u - \nabla p, \\ \partial_t^2 u &= \nu \Delta \partial_t u - (\partial_t u \cdot \nabla) u - (u \cdot \nabla) \partial_t u - \nabla \partial_t p, \\ \partial_t^3 u &= \nu \Delta \partial_t^2 u - (\partial_t^2 u \cdot \nabla) u - 2(\partial_t u \cdot \nabla) \partial_t u \\ &\quad - (u \cdot \nabla) \partial_t^2 u - \nabla \partial_t^2 p. \end{aligned}$$

The complete pattern is

$$\partial_t^{r+1} u = \nu \Delta \partial_t^r u - \sum_{a=0}^r \binom{r}{a} (\partial_t^a u \cdot \nabla) \partial_t^{r-a} u - \nabla \partial_t^r p.$$

Only now do we compress the notation:

$$V_r := \partial_t^r u, \quad Q_r := \partial_t^r p.$$

Pressure has to travel with every response. The compact tower is

$$Q_r = R_i R_j \sum_{a=0}^r \binom{r}{a} V_{a,i} V_{r-a,j},$$

$$V_{r+1} = \nu \Delta V_r - \sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} - \nabla Q_r.$$

Now the connection to spatial smoothness is visible. A hypothetical singularity runs forward through rougher and faster temporal responses. The proof reads the viscous relation backward. Once a finite response and its transport–pressure terms are controlled, the equation isolates ΔV_r and returns two spatial derivatives to the preceding response. Each finite climb in time opens a finite climb in Sobolev height when traced back to the original velocity. As the temporal order rises, the Sobolev height available to u rises with it.

Now return to the critical height that scaling exposed. Let

$$\Lambda := (-\Delta)^{1/2}, \quad E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2, \quad H(t) := E_{1/2}(t).$$

Pairing Navier–Stokes at height s gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1},$$

where $\mathcal{P}_s$ is the complete signed nonlinear production at that height. Repeated differentiation at $s = \frac{1}{2}$ now gives

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2},$$

$$H'' = \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2},$$

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

At every finite order,

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

The n -th temporal derivative has exposed the spatial energy at height $n + \frac{1}{2}$. The calculation lifts directly to every temporal response:

$$E_{r,s}(t) := \frac{1}{2} \|\Lambda^s V_r(t)\|_2^2, \quad \partial_t E_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}.$$

If every finite height is controlled, then any requested classical derivative follows by choosing n high enough:

$$H^{n+1/2}(\mathbb{R}^3) \hookrightarrow C_b^k(\mathbb{R}^3) \quad \text{whenever } n > k + 1,$$

and the full spatial intersection gives

$$\bigcap_{m < \infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

This is the realization behind the tower. It is not yet the bound. The differentiated identities show which spatial coordinates have to be paid for; they do not make those coordinates finite by rearrangement.

The payment is the whole-terminal rectangle theorem. Fix any finite temporal depth N and spatial height M . For each $0 \leq r \leq N$, it supplies

$$V_r \in L^2(0, T_*; H^{M+1}), \quad V_{r+1} \in L^2(0, T_*; H^{M-1}),$$

together with the pressure coordinates carried by the recurrence. The Hilbert-pair trace step then gives

$$V_r \in C([0, T_*]; H^M).$$

This is one finite rectangle. Its constant may depend on N and M . The proof never asks one constant to control infinitely many derivatives at once. It proves the required statement for every finite rectangle.

Those rectangles are not unrelated solutions. Every coordinate belongs to one velocity history generated by u_0 . When two rectangles overlap, their shared coordinates agree. Compatibility lets us assemble the entire family:

$$u_0 \longrightarrow \{\mathcal{R}_{N,M}[u_0; T_*]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] := \bigcap_{r,m < \infty} H^r(0, T_*; H^m(\mathbb{R}^3)), \quad u \in \mathcal{I}[u_0].$$

No infinite-order estimate has been smuggled into this step. Each coordinate came from a finite theorem; the intersection records that all of them belong to one compatible fluid.

Now take the zeroth temporal row. For every finite m , the rectangle pair and the equation place

$$u \in C([0, T_*]; H_\sigma^m).$$

All of these traces meet at one terminal value,

$$u_* := u(T_*) \in \bigcap_{m < \infty} H_\sigma^m = H_\sigma^\infty \hookrightarrow C^\infty.$$

At T_* , the equation converts the completed spatial information back into time information. Starting from $V_0^* = u_*$, the pressure formula gives Q_0^* , and the recurrence gives V_1^* . Repeating this at every finite order produces the pressure-complete terminal jet. Smooth local existence restarts from u_* ; uniqueness joins that continuation to the original history. This extends the solution past T_* , contradicting finite maximality. Hence

$$T_* = \infty.$$

Once the intersection is present, the familiar estimates are read from it instead of proved one by one. Given a finite response order r , a spatial derivative order k , and $2 \leq p \leq \infty$, choose

$$m > k + \frac{3}{2} - \frac{3}{p}.$$

That finite coordinate gives

$$\mathcal{I}[u_0] \longrightarrow H_t^1 H_x^m \longrightarrow C_t H_x^m \longrightarrow L_t^q W_x^{k,p} \quad (1 \leq q \leq \infty).$$

Choosing $m = 1, 2, 3$ in turn recovers, among many others,

$$L_t^\infty L_x^6, \quad L_t^\infty L_x^\infty, \quad L_t^\infty W_x^{1,\infty}.$$

These are not separate achievements. Each is one finite readout of $\mathcal{I}[u_0]$.

The whole motion is

$$u_0 \longrightarrow \{(V_r, Q_r)\}_{r \geq 0} \longrightarrow \{\mathcal{R}_{N,M}[u_0]\}_{N,M < \infty} \longrightarrow \mathcal{I}[u_0] \longrightarrow u_* \in H^\infty \longrightarrow \text{restart} \longrightarrow T_* = \infty.$$